

Phoenix Cyber Resilience Project

CY021 – Secure Document Management System

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1. EXECUTIVE SUMMARY

The Secure Document Management System developed for the Phoenix Cyber Resilience Project provides a structured and secure framework for managing governance, compliance, operational, and cybersecurity-related documentation. As organisations increasingly rely on digital collaboration and cloud-based storage, secure document management becomes essential for protecting sensitive information and maintaining operational integrity.

This framework proposes the implementation of a centralised documentation environment supported by role-based access controls, encryption, version tracking, backup procedures, and governance-focused security controls. The proposed system aligns with recognised cybersecurity standards and aims to improve confidentiality, accountability, collaboration, and cyber resilience across the Phoenix project environment.

2. INTRODUCTION

Documentation plays a critical role in cybersecurity governance, operational management, and compliance activities. Organisations rely on documentation systems to manage policies, procedures, incident reports, risk assessments, audit records, and technical configurations. Without secure document management practices, sensitive information may become vulnerable to unauthorised access, modification, or accidental loss.

This document presents the proposed Secure Document Management System for the Phoenix Cyber Resilience Project. The framework focuses on improving document security, access management, governance oversight, and collaboration processes while supporting long-term cyber resilience objectives.

The proposed solution also supports compliance with recognised cybersecurity frameworks and industry best practices.

3. PURPOSE OF THE DOCUMENT MANAGEMENT SYSTEM

The primary purpose of the Secure Document Management System is to provide a centralised, secure, and organised environment for storing and managing project documentation.

Key Objectives

- Protect sensitive governance and cybersecurity documents
- Improve document accessibility for authorised users
- Maintain accountability through audit logging
- Prevent unauthorised modifications or disclosures
- Support collaboration across project teams
- Improve version control and document tracking
- Support compliance and governance requirements

The system also aims to reduce operational risks associated with fragmented storage methods, inconsistent document handling, and poor access management practices.

4. PROPOSED SYSTEM ARCHITECTURE

The proposed architecture uses a secure centralised repository supported by authentication mechanisms, encryption controls, and monitoring systems.

Proposed Architecture Flow

Users



Authentication & MFA



Role-Based Access Control



Secure Document Repository



Encrypted Backup Storage

Core Components

Component	Purpose
Central Repository	Secure document storage
Authentication Server	User verification
Access Control System	Permission management
Audit Logging System	Activity monitoring
Backup Server	Disaster recovery support

The architecture is designed to improve both security and operational efficiency across governance and compliance activities.

5. DOCUMENT CLASSIFICATION STRUCTURE

Documents within the Phoenix project should be classified based on sensitivity and operational importance.

Classification	Description	Example
Public	Non-sensitive documents	Public project overview
Internal	Internal team documents	Meeting notes
Confidential	Sensitive operational files	Risk assessments
Restricted	Highly sensitive documents	Credentials and security keys

Restricted documents require enhanced security controls including encryption, limited access permissions, and detailed activity monitoring.

6. ACCESS CONTROL & AUTHENTICATION

Strong access management controls are essential for preventing unauthorised document access and maintaining information confidentiality.

Recommended Security Controls

- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Principle of Least Privilege
- Password complexity requirements
- Account lockout mechanisms
- Session timeout policies

Example Access Levels

User Role	Access Level
Governance Lead	Full Access
Security Analyst	Limited Operational Access
Team Members	Assigned Document Access
External Users	Restricted Access

These controls help minimise insider threats, accidental disclosures, and privilege misuse.

7. VERSION CONTROL & DOCUMENT TRACKING

Version control mechanisms improve accountability and document integrity by tracking changes made to files over time.

Version Control Features

- Change tracking
- Revision history
- Document approval workflows
- Rollback capability
- Contributor identification

Example Version Table

Version	Description	Author	Date
v1.0	Initial document creation	Dilano	May 2026
v1.1	Updated security controls	Dilano	May 2026

Version tracking improves collaboration and reduces confusion caused by duplicate or outdated documents.

8. SECURITY CONTROLS & PROTECTION MECHANISMS

The proposed system includes multiple layers of security controls to improve protection against cyber threats and unauthorised access.

Recommended Protection Mechanisms

Security Control	Purpose
AES-256 Encryption	Protect stored data
TLS/HTTPS	Secure data transmission
Activity Logging	Monitor document access
File Integrity Monitoring	Detect unauthorised modifications
Backup Encryption	Protect recovery data

Security monitoring should also include regular reviews of access logs and suspicious activity detection.

9. BACKUP & RECOVERY STRATEGY

Backup procedures are necessary to maintain operational continuity during cyber incidents, accidental deletion, or system failures.

Proposed Backup Strategy

- Daily incremental backups
- Weekly full-system backups
- Cloud and offline redundancy
- Backup encryption
- Periodic restoration testing

The system should also include a Disaster Recovery Plan (DRP) outlining recovery priorities and restoration procedures.

10. DOCUMENT LIFECYCLE MANAGEMENT

Proper lifecycle management ensures documents remain accurate, secure, and properly maintained throughout their operational use.

Document Lifecycle Stages

1. Document Creation
2. Review & Approval
3. Secure Storage
4. Controlled Distribution
5. Archiving
6. Secure Disposal

Lifecycle Management Benefits

- Improved organisation
- Reduced duplication
- Better compliance management
- Improved accountability
- Reduced storage risks

11. COMPLIANCE & GOVERNANCE ALIGNMENT

The proposed document management framework aligns with recognized cybersecurity and governance standards.

Relevant Frameworks

Framework	Purpose
ISO/IEC 27001	Information security management
NIST Cybersecurity Framework	Risk and governance guidance
ACSC Essential Eight	Security mitigation strategies
Australian Privacy Principles	Privacy compliance

Compliance alignment supports governance maturity and improves operational credibility.

12. BENEFITS OF THE PROPOSED SYSTEM

The Secure Document Management System provides several operational and governance-related advantages.

Key Benefits

- Improved document security
- Better collaboration across teams
- Enhanced accountability and traceability
- Reduced risk of information loss
- Faster document retrieval
- Improved compliance readiness
- Stronger cyber resilience capabilities

The implementation of centralized governance controls also improves long-term operational efficiency.

13. RECOMMENDATIONS

The following recommendations are proposed to strengthen document management security within the Phoenix project:

- Implement mandatory multi-factor authentication
- Conduct regular access permission reviews
- Introduce automated backup monitoring
- Perform periodic compliance audits
- Maintain detailed activity logging
- Apply encryption to all sensitive documents
- Provide cybersecurity awareness training

These improvements will support secure governance operations and reduce exposure to cybersecurity risks.

14. CONCLUSION

The Secure Document Management System developed for the Phoenix Cyber Resilience Project provides a structured and secure approach for managing governance, compliance, and cybersecurity-related documentation.

Through the implementation of strong access controls, encryption mechanisms, version tracking, backup strategies, and governance-focused policies, the framework improves confidentiality, integrity, accountability, and operational resilience.

The proposed system supports secure collaboration while aligning with recognized cybersecurity standards and best practices. Continuous monitoring and ongoing improvements will remain essential for maintaining effective document security within the evolving cyber threat landscape.